Balaji Madhan

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Summary

Computer Science undergraduate specializing in Artificial Intelligence and Machine Learning, with hands-on experience building full-stack AI systems using RAG pipelines, vector search, and LLMs. Skilled in Python-based ML/DL development, from data preprocessing to model deployment, with multiple internship experiences applying machine learning to real-world enterprise problems.

Education

SRM Institute of Science and Technology, Trichy **June 2023 – June 2027**

Bachelor of Technology in Computer Science (AI & ML Specialization)

- **CGPA: 8.68 / 10**

Projects

RepoLens – AI-Powered Repository Intelligence Platform

- Developed a full-stack AI platform that clones and analyzes GitHub repositories using RAG, Qdrant vector search, and Gemini AI.
- Implemented semantic code search, dependency graph visualization, repository summarization, and AI-powered codebase assistance using FastAPI and Next.js.
- Built and verified an end-to-end pipeline from repository ingestion and embedding generation to grounded AI responses for intelligent code understanding.

Alzheimer Disease Prediction System | *Python, Machine Learning, Deep Learning, RAG*

- Developed an end-to-end AI system for early Alzheimer's disease risk prediction using ensemble Machine Learning and Deep Learning models.
- Integrated RAG pipelines with vector search (FAISS/ChromaDB) and LLMs to retrieve relevant medical literature and generate explainable clinical recommendations.

AI Resume Screening System

- Developed an intelligent recruitment platform that automatically analyzes resumes, extracts skills using NLP, and ranks candidates against job descriptions.
- Integrated RAG and LLMs to generate candidate summaries and skill-gap analysis using semantic search and vector databases.

Technical Skills

Languages: Python, SQL, Fullstack, Blockchain, VLSI Development, Oracle Database, C, C++, CUDA
ML & Data Tools: NumPy, Pandas, Scikit-learn, Matplotlib, TensorFlow (basic)
Frameworks & Platforms: Jupyter Notebook, Google Colab, Git, GitHub
Databases: MySQL, SQLite, HeidiSQL, Oracle Database
Concepts: Regression, Classification, Feature Engineering, Model Evaluation
Other: Linux Basics, Agile Methodologies, Technical Writing, Aptitude Training
Relevant Coursework: Data Structures, Machine Learning, Algorithms Analysis, Database Management, Operating Systems, Computer Architecture, Software Engineering, Agile and Scrum Methodologies, Software Development Lifecycle (SDLC), Engineering Mathematics, Accounting

Internship Experience

KrypC Technologies Pvt Ltd **May 2025 – July 2025**

Artificial Intelligence Intern (On-Site)

- Interned at KrypC Technologies Pvt Ltd, contributing to AI-based solutions for enterprise use cases.
- Worked on real-time machine learning models, handling data collection, preprocessing, and feature engineering for production-grade pipelines.
- Collaborated closely with mentors and senior engineers to design, build, and optimize AI algorithms using Python and relevant ML frameworks.
- Participated in the complete AI development lifecycle – from data preprocessing and model training to evaluation, deployment, and documentation.

Prodigy Infotech **December 2024**

Machine Learning Intern

- Worked on real-time predictive modeling projects using supervised learning techniques.
- Implemented ML algorithms for classification and regression using Python and Scikit-learn.
- Gained hands-on experience in data preprocessing, model evaluation, and visualization.

Skillcraft Technologies **March 2025**

Machine Learning & Graphic Design Intern

- Built machine learning models and supported the development of data-driven features.
- Designed professional-level graphics, banners, and social media creatives using Canva and Adobe tools.